**Client Change Request (CCR)**

**CCR Number:** CCR-013

**Client:** Data Analytics Solutions, Inc.

**Submitted By:** Thomas Reynolds, Senior Data Analyst

**Priority:** Medium

**Requested Delivery:** End of Today's Sprint

**Assigned Developer:** _____

**Date Assigned:** _____

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**Business Background**

Data Analytics Solutions processes large amounts of numerical data collected from sensors located throughout manufacturing facilities. As the amount of data has increased, the current software has become slower because one program performs every task sequentially.

To improve performance, the company would like to begin dividing work among multiple processes. One process should be responsible for generating a sequence of numbers, while a second process should receive those numbers and calculate summary statistics. This approach will allow future versions of the software to process larger data sets more efficiently.

For this initial prototype, only two processes are required.

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**Customer Requirements**

The program should divide the work into two independent processes.

The first process should:

- Generate a sequence of numbers.
- Send those numbers to the second process.
- Display a message when it begins processing.
- Display a message when it has completed its work.

The second process should:

- Receive the generated numbers.
- Calculate summary statistics for the received data.
- Display a message when it begins processing.
- Display a message when it has completed its work.

After both processes have completed, the program should display the calculated statistics in a clear and organized report.

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**Acceptance Criteria**

The customer will consider the software complete when all of the following are true.

□ The program creates two separate processes.

□ The first process generates a sequence of numbers.

□ The generated numbers are successfully transferred to the second process.

□ The second process correctly receives the data.

□ The second process calculates summary statistics.

□ Both processes display a message when they start.

□ Both processes display a message when they finish.

□ The calculated statistics are displayed in a readable report.

□ The program terminates normally after both processes have completed.

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**Out of Scope**

The customer is **not** requesting:

- More than two processes
- Reading numbers from a file
- Saving results to a file
- Processing multiple data sets
- Network communication
- Threads or multithreading
- Graphical user interfaces (GUI)
- Error logging
- Performance benchmarking
- Distributed computing

If these features appear in the software, they will not be evaluated.

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**Example Scenario 1**

**Program Execution**

./statistics

**Expected Output**

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Data Analytics Solutions

Statistics Processing

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Parent Process Started

Child Process Started

Generating Numbers...

Numbers Sent Successfully

Receiving Numbers...

Calculating Statistics...

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Count  : 10

Minimum : 5

Maximum : 42

Average : 21.80

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Child Process Finished

Parent Process Finished

**Example Scenario 2**

**Program Execution**

./statistics

**Expected Output**

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Data Analytics Solutions

Statistics Processing

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Parent Process Started

Generating Numbers...

Child Process Started

Receiving Numbers...

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Count  : 5

Minimum : 10

Maximum : 50

Average : 30.00

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Child Process Finished

Parent Process Finished

**Note:** Because the program uses two independent processes, the exact ordering of some status messages may vary slightly depending on the operating system's scheduler. The required functionality—not the exact ordering of every message—will be used for grading.

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### Notes from the Customer

"We're beginning to redesign our software so that multiple parts of the program can work simultaneously. For this first version, I simply want one process to generate data and another process to analyze it. I also want employees to know when each process begins and when it finishes so they can better understand what the software is doing."

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### Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

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### Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

1.

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**Reminder**

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 15**, including:

- Variables and constants
- Input/output
- Functions
- Pointers
- Strings
- Arrays
- Command-line arguments
- Process creation using fork()
- Interprocess communication using pipes
- Basic process synchronization (wait())

Do **not** use C language features that have not yet been introduced in the course.